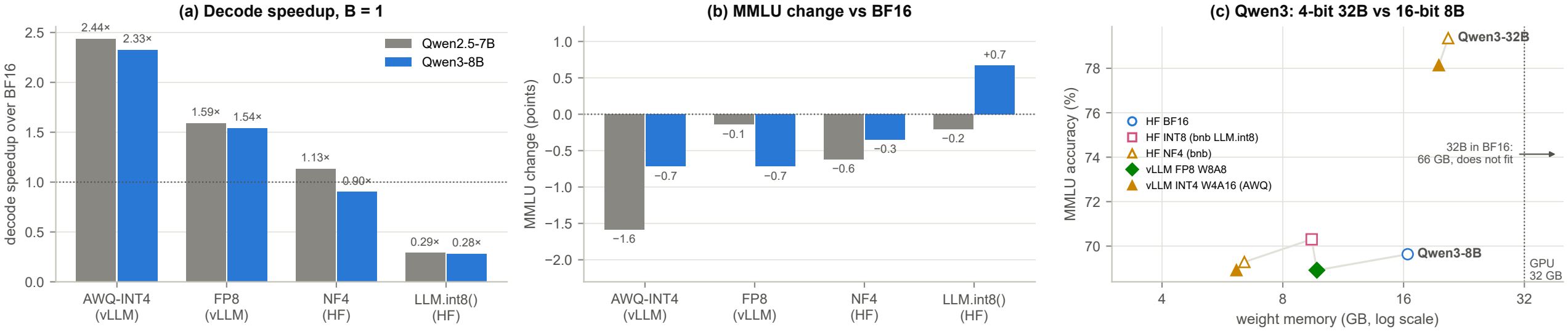
(a) Decode speedup, B = 1
decode speedup over BF16
2.5
2.0
1.5
1.0
0.5
0.0
2.44×
2.33×
1.59×
1.54×
1.13×
0.90×
0.29×
0.28×
Qwen2.5-7B
Qwen3-8B
AWQ-INT4 (vLLM)
FP8 (vLLM)
NF4 (HF)
LLM.int8() (HF)
(b) MMLU change vs BF16
MMLU change (points)
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1.6
−0.7
−0.1
−0.7
−0.6
−0.3
−0.2
+0.7
AWQ-INT4 (vLLM)
FP8 (vLLM)
NF4 (HF)
LLM.int8() (HF)
(c) Qwen3: 4-bit 32B vs 16-bit 8B
MMLU accuracy (%)
78
76
74
72
70
Qwen3-32B
HF BF16
HF INT8 (bnb LLM.int8)
HF NF4 (bnb)
vLLM FP8 W8A8
vLLM INT4 W4A16 (AWQ)
32B in BF16:
66 GB, does not fit
Qwen3-8B
GPU
32 GB
4
8
16
32
weight memory (GB, log scale)